Fractional directed-cycle packing at least half the minimum outdegree

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Editorial note. Complete original active mathematical body and embedded bibliography retained. Authors own mathematical correctness; no proof repair, extra hypothesis, independent counterexample search or certification.

Editorial note. Original AI-assistance disclosure989–991 retained exactly: AI-assisted refinement improved the explicit asymptotic error and aided the stopped/frozen Z_j(v) sequence construction. No inference of an AI-discovered core proof or pure-human/AI-free claim is made.

Editorial note. The main asymptotic theorem keeps absolute constants and k-threshold uniform over every graph order n; the separate fractional packing theorem is not conflated with integral packing. Direct local/epsilon consequences and auxiliary lemmas are not separately counted.

Editorial note. Native inactive draft material remains in the raw source; no discarded proof is substituted for the active author proof.

Editorial note. Rendering-only adaptation at original line 951: use the native thm-restate starred MainThm restatement, retaining the original stored theorem and its thm:main target at lines 132–139. The original unstarred repetition re-registered the same theorem and labels; raw author source and failed r001 are preserved. No mathematical text or reference target is changed.

Allowed standard prerequisites (editorial).

- Directed cycle packings, fractional feedback vertex sets, LP duality and martingale concentration. Native finite digraph/outdegree conventions, fractional packing/dual definitions, all minimal-counterexample reductions, probability estimates, stopped/frozen sequence construction and explicit final constants retained.. Full active original mathematical paper and embedded bibliography retained, with all nonadjacent core arguments bundled into the corresponding target.

1 Introduction

Throughout, digraphs are finite and simple; pairs of opposite arcs are allowed, and every cycle is directed and has length at least two. We write $\delta^+(D)$ for the minimum out-degree of a digraph D and generally follow the notation and terminology of [3].

Finding disjoint cycles under degree conditions is a classical problem in extremal graph theory. In the undirected setting, the Corrádi–Hajnal Theorem [8] asserts that every graph with at least $3k$ vertices and minimum degree at least $2k$ contains k disjoint cycles. The directed analogue is considerably more delicate. The transitive tournament has maximum possible total degree at every vertex, yet contains no directed cycle; thus no condition on the total degree alone can force a directed cycle, and one must impose a lower bound on the out-degree. This is the natural directed counterpart of the undirected condition: every finite digraph with positive minimum out-degree contains a directed cycle. The central question is how large the minimum out-degree must be to guarantee a prescribed number of disjoint directed cycles.

For a positive integer k, let $f(k)$ be the least integer d such that every digraph with minimum out-degree at least d contains k disjoint directed cycles. In 1981, Bermond and Thomassen proposed the following conjecture.

Conjecture 1.1 (Bermond–Thomassen Conjecture [5])**.** *For every positive integer k, every digraph with minimum out-degree at least $2k-1$ contains k disjoint directed cycles.*

The threshold $2k-1$ is best possible. Indeed, the complete symmetric digraph on $2k-1$ vertices has minimum out-degree $2k-2$, while it contains at most $k-1$ disjoint directed cycles; hence $f(k) \geq 2k-1$.

Thomassen [14] proved in 1983 that $f(k)$ is finite for every k, obtaining $f(k) \leq (k+1)!$, and settled the conjecture for $k=2$. Alon [1] then introduced a probabilistic approach and showed that $f(k) \leq 64k$, establishing that the required out-degree grows linearly in the number of cycles. Bucić [7] refined this approach–incorporating the known case of three disjoint cycles–and improved the bound to $f(k) \leq 18k$. While these results determine the correct order of growth, they leave a wide gap to the conjectured threshold.

Progress on the exact conjecture has followed a complementary path. The case $k=1$ is immediate; after Thomassen's elegant proof for the case $k=2$, the case $k=3$ was proved by Lichiardopol, Pór, and Sereni [11]. Among tournaments, Bang-Jensen, Bessy, and Thomassé [4] showed that minimum out-degree at least $2k-1$ guarantees k disjoint directed triangles. More recently, Bessy, Muis, Sereni, Steiner, and Wiederrecht [6] proved that the conjectured degree condition guarantees k disjoint near-directed cycles, that is, oriented cycles that can be made directed by flipping the direction of at most one arc. The exact conjecture for general digraphs remains open for $k \geq 4$.

In this paper we determine the asymptotic value of $f(k)$, resolving the Bermond–Thomassen conjecture to leading order with an explicit sublinear error term.

Theorem 1.2. *There exist absolute constants $C > 0$ and K such that, for all integers $k \geq K$ and $n \geq 1$, every n-vertex digraph D with*

$$\delta^+(D) \geq 2k + Ck^{3/4}\sqrt{\log k}$$

contains k disjoint directed cycles.

In fact, our method reaches the explicit choice $K = 10^{12}$ and $C = 64\sqrt{2}$, see Section 6. Together with the complete symmetric digraph construction, $2k-1 \leq f(k) \leq 2k+O\big(k^{3/4}\sqrt{\log k}\big)$, so $\lim_{k\to\infty} f(k)/k = 2$. Equivalently, for every $\varepsilon > 0$, minimum out-degree at least $(2+\varepsilon)k$ guarantees k disjoint directed cycles once k is sufficiently large. Moreover, by Lemma 3.5, the condition $k \geq K$ cannot be replaced by an assumption of sufficiently large order n if we wish to retain the same minimum out-degree bound. Lemma 3.5 is formally stated and proved in Section 3.

The rest of the paper is organized as follows. We begin with Section 2, which contains some notation and terminology. In Section 3 we recall fractional cycle packings and state the standard probabilistic estimates used later. Section 4 bounds the order of a minimum counterexample by means of contractions and an independent-set argument. In Section 5 we prove a lower bound on the fractional cycle packing number (Theorem 5.2), construct a cycle distribution with bounded vertex incidence (Lemma 5.3), and show that cycles can be deleted successively so that a suitable potential function does not increase (Lemma 5.4). Finally, in Section 6 we combine these ingredients to prove the main theorem.

2 Notation

For a positive integer a, let $[a] := \{1, 2, \ldots, a\}$; for integers $a < b$, let $[a, b] := \{a, a+1, \ldots, b\}$. We write $x \ll y$ when x is sufficiently small as a function of y. The notation $t = a \pm b$ means $a - b \leq t \leq a + b$. All logarithms are natural.

Let $D = (V, E)$ be a digraph. The order of D is the number of vertices in D, denoted by $|V(D)|$ or $|D|$. We will denote an edge oriented from u to v by uv. We use $D[X]$ to denote the subdigraph induced by a vertex set X. Let $D - X = D[V \setminus X]$. For $v \in V(D)$, we denote the set of out-neighbors and in-neighbors of v in D by $N^+(v)$ and $N^-(v)$ respectively, and we write $d^+(v) = |N^+(v)|$ and $d^-(v) = |N^-(v)|$ for the out-degree and in-degree of v, respectively. For $S \subseteq V(D)$, we write $N^-(v, S) = N^-(v) \cap S$, $N^+(v, S) = N^+(v) \cap S$. Set $d^-(v, S) = |N^-(v, S)|$ and $d^+(v, S) = |N^+(v, S)|$. Write $\nu(D)$ for the maximum number of disjoint cycles in D. A digraph $D = (V, E)$ is strongly connected (or just strong) if it has a directed path from x to y for every ordered pair x, y of vertices.

3 Preliminaries

3.1 Fractional cycle packings

Fractional cycle packings with vertex capacity one were studied by Seymour [13], who related their value to the size of a minimum feedback vertex set. We require a complementary estimate: a lower bound on

the fractional packing number in terms of the minimum out-degree, which we prove in this section. Let $\mathcal{C}(G)$ be the set of cycles of G. A *fractional cycle packing* assigns a number $x_C \geq 0$ to each $C \in \mathcal{C}(G)$ so that $\sum_{C \ni v} x_C \leq 1$ for every vertex v. Its total weight is $\sum_C x_C$, and the largest possible total weight is denoted by $\nu^*(G)$. If G has no cycle, we set $\nu^*(G) = 0$.

The dual object is a *fractional feedback vertex set*: it assigns weights $y_v \geq 0$ to vertices so that $\sum_{v \in V(C)} y_v \geq 1$ for every cycle C. Write $y(X) = \sum_{v \in X} y_v$ for a set of vertices X. Both problems are feasible: the primal by setting $x_C = 0$ for every cycle, and the dual by setting $y_v = 1$ for every vertex. By finite linear programming duality [12, Chapter 7],

$$\nu^*(G) = \min\{y(V(G)) : y \text{ is a fractional feedback vertex set}\}. \tag{3.1}$$

Both optima are attained: the primal feasible region is compact since $0 \leq x_C \leq 1$, and in the dual any weight $y_v > 1$ can be replaced by 1 without violating a cycle constraint. We may therefore choose an optimal dual assignment with $0 \leq y_v \leq 1$.

3.2 Standard estimates

We will use the following standard estimates.

Lemma 3.1 (Chernoff bound)**.** *Let $X \sim \mathrm{Bin}(n,p)$ with mean $\mu = np$. Then for $0 < \tau < 1$,*

$$\mathbb{P}\big[|X - \mu| > \tau\mu\big] \leq 2e^{-\tau^2\mu/3}.$$

Lemma 3.2 (Markov's inequality)**.** *If $X \geq 0$ is a random variable, then for every $t > 0$,*

$$\mathbb{P}[X \geq t] \leq \frac{\mathbb{E}[X]}{t}.$$

Lemma 3.3 (Bernoulli's inequality)**.** *For every integer $q \geq 1$ and every real $x \leq 1$,*

$$(1-x)^q \geq 1 - qx.$$

In particular, if $0 \leq x < 1/(q+1)$, then $(1-x)^{q+1} \geq 1 - (q+1)x > 0$.

Lemma 3.4. *For $q \geq 1$ and $0 \leq x < 1/(q+1)$,*

$$(1-x)^{-q} \leq 1 + \frac{qx}{1-(q+1)x}.$$

Proof. For $0 \leq x < 1/(q+1)$, apply the mean value theorem to $f(t) = (1-t)^{-q}$ on $[0,x]$. There exists $c \in (0,x)$ such that $(1-x)^{-q} - 1 = qx(1-c)^{-q-1}$. Since $0 < c < x$, we have $(1-c)^{-q-1} \leq (1-x)^{-q-1}$, so $(1-x)^{-q} - 1 \leq qx(1-x)^{-q-1}$. Bernoulli's inequality gives $(1-x)^{q+1} \geq 1-(q+1)x > 0$, and therefore $(1-x)^{-q-1} \leq 1/(1-(q+1)x)$. Combining these two estimates, the claim follows. □

3.3 Equivalence properties

In this subsection, we establish an equivalence principle that allows us to reduce general cycle packing problems in digraphs to the setting of sufficiently large strong out-regular digraphs.

Lemma 3.5. *Fix integers $k \geq 1$ and $d \geq 1$. The following statements are equivalent.*

(i) *Every finite simple loopless digraph D satisfying $\delta^+(D) \geq d$ contains k disjoint directed cycles.*

(ii) *There exists an integer $n_0 = n_0(k,d)$ such that every finite simple loopless digraph D satisfying $|V(D)| \geq n_0$ and $\delta^+(D) \geq d$ contains k disjoint directed cycles.*

(iii) *There exists an integer $n_0 = n_0(k,d)$ such that every strong d-out-regular digraph D satisfying $|V(D)| \geq n_0$ contains k disjoint directed cycles.*

The same equivalence holds if all digraphs above are required to be oriented graphs, that is 2-cycles are not allowed.

Proof. The implications (i) $\Longrightarrow$ (ii) $\Longrightarrow$ (iii) follow directly from definitions. We prove that (iii) $\Longrightarrow$ (i). Assume statement (i) does not hold. Then there exists a finite simple loopless digraph D such that $\delta^+(D) \geq d$ and $\nu(D) < k$.

First retain exactly d outgoing arcs at every vertex of D, deleting all other arcs, and denote the resulting spanning subdigraph by D'. Then $d^+_{D'}(v) = d$ for all $v \in V(D')$, while deleting arcs cannot create new directed cycles, and hence $\nu(D') \leq \nu(D) < k$. The strong components of D' admit an acyclic ordering. Let H be a terminal strong component in this ordering. Since no arc of D' leaves H, every vertex of H has all its d outgoing arcs inside H. Consequently H is strong and d-out-regular, and $\nu(H) \leq \nu(D') < k$. As $d \geq 1$, the digraph H contains an arc.

We now describe an order increasing operation. Let G be any finite simple loopless digraph with an arc uv, and let $m \geq 1$. Put $A = N^+_G(u) \setminus \{v\}$. Introduce new vertices $x_1, \ldots, x_m$, delete uv, keep every other old arc, and add $ux_1,\ x_1x_2, \ldots,\ x_{m-1}x_m,\ x_mv$, together with x_iw for $i \in [m],\ w \in A$. Denote the resulting digraph by G_m. Every old vertex keeps its out-degree. Moreover, $d^+_{G_m}(x_i) = d^+_G(u)$ for $i \in [m]$, and therefore $\delta^+(G_m) = \delta^+(G)$.

We claim that $\nu(G_m) = \nu(G)$. Indeed, every directed cycle of G_m meeting a new vertex must contain u. Such a cycle contains a consecutive segment $ux_1x_2 \cdots x_iw$ for some $w \in N^+_G(u)$. Replacing this segment by the old arc uw gives a directed cycle of G with exactly the same old vertices. Since every cycle meeting the new vertices contains u, at most one member of a disjoint cycle family can meet the new vertices. Thus an entire packing in G_m can be compressed simultaneously to a packing of the same cardinality in G, proving $\nu(G_m) \leq \nu(G)$. Conversely, take a maximum packing in G. At most one of its members uses the arc uv. If such a member exists, replace its occurrence of uv by $ux_1x_2 \cdots x_mv$. All other members remain unchanged, and the inserted vertices are new, so disjointness is preserved. Hence $\nu(G_m) \geq \nu(G)$.

The same construction preserves strong connectivity: every old directed path can be lifted by replacing uv, when necessary, by the new path, while every new vertex is reachable from u and can reach v. By construction, the new digraph G_m is an oriented graph whenever G is an oriented graph. Applying the operation above to an arc of H, we obtain, for every integer $N \geq |V(H)|$, a strong d-out-regular digraph H_N of order exactly N with $\nu(H_N) = \nu(H) < k$. This produces counterexamples to statement (iii) with arbitrarily large order, contradicting statement (iii). Hence (iii) $\Rightarrow$ (i). This proves the equivalence of (i)–(iii). □

As an additional consequence of the construction, a single finite counterexample automatically produces counterexamples of every larger order.

4 The order of a minimum counterexample

Fix integers $k \geq 2$ and $r \geq 2k$. In this section, a *counterexample* is a digraph D with $\delta^+(D) \geq r$ and $\nu(D) < k$. The parameters r and k remain fixed whenever minimum order is used.

The contraction argument in Lemma 4.2 and the independent set count in Proposition 4.3 follow Alon [1, Claims 1 and 2]; see also Bucić [7, Propositions 3.1 and 3.2]. We give the arguments for the fixed pair (r, k) used here.

Lemma 4.1 justifies contractions that create loops, without an assumption on $f(k-1)$. Contraction may create a loop. The next lemma allows us to use such a contraction without changing the minimum possible order of a counterexample. For a digraph G in which loops are allowed, count a loop once in the out-degree and regard it as a cycle of length one. Write $\nu_L(G)$ for the maximum number of disjoint cycles under this convention.

Lemma 4.1. *If a loopless counterexample exists, its minimum possible order equals the minimum possible order of a counterexample in which loops are allowed.*

Proof. Recall that a counterexample is a digraph D with $\delta^+(D) \geq r$ and $\nu(D) < k$. Write m_0 for the minimum order of a loopless counterexample and m_L for the minimum order of a counterexample in which loops are allowed. Thus it suffices to show $m_0 = m_L$. The inequality $m_L \leq m_0$ is immediate, since every loopless digraph is also a digraph in which loops are allowed.

It remains to prove $m_0 \leq m_L$. Let L be the set of looped vertices of a digraph G. Then $\nu_L(G) = |L| + \nu(G - L)$. Choose a counterexample of minimum order m_L among digraphs in which loops are allowed, and, subject to this, with as many arcs as possible. We shall use this counterexample to build a loopless counterexample of the same order, which will give $m_0 \leq m_L$.

We claim that every looped vertex is adjacent in both directions to every other vertex. Otherwise, if some $v \in L$ has no arc to w with $w \neq v$, adding that arc would leave $\nu_L(G) = |L| + \nu(G - L)$ unchanged, since neither L nor $G - L$ is affected, while preserving the out-degree condition, contradicting the choice of G.

Since every out-degree is at most m_L, we have $m_L \geq r$. Equality $m_L = r$ would force every possible arc and every loop to be present, giving $\nu_L(G) = r \geq k$, contrary to G being a counterexample. Thus $m_L \geq r+1$. Now delete all loops of G, retaining every vertex and every arc between distinct vertices. A formerly looped vertex has out-degree $m_L - 1 \geq r$, while every other out-degree is unchanged. Deleting arcs cannot create a family of k disjoint cycles. The resulting digraph is therefore a loopless counterexample of order m_L, so $m_0 \leq m_L$ and this proves the claim. □

The contraction argument below follows Thomassen [14] (also can be found in Alon [1, Claims 1 and 2]); we restate it for the fixed pair (r, k) used here.

Lemma 4.2. *If a counterexample exists, then there exists a counterexample of minimum order that is r-out-regular. In any such digraph D, the endvertices of every arc have a common in-neighbour, and $D[N^-(v)]$ contains a cycle for every vertex v.*

Proof. Choose a counterexample of minimum order, and retain exactly r outgoing arcs at each vertex. The resulting spanning subdigraph D is still a counterexample of the same order. By Lemma 4.1, D is also of minimum order among counterexamples in which loops are allowed.

We first show that the endvertices of every arc have a common in-neighbour. Suppose, for a contradiction, that an arc uv has no common in-neighbour. Delete all outgoing arcs of u except uv, and identify u and v to a single vertex z. The outgoing arcs of z are those inherited from v; we retain a loop at z precisely when vu is also an arc of D, and discard the loop arising solely from uv. Parallel arcs are identified. Since v has r distinct out-neighbours, z has r distinct outgoing arcs, counting a possible loop. By the assumption that uv has no common in-neighbour, no vertex outside $\{u, v\}$ loses an out-neighbour. Hence the resulting digraph D' has minimum out-degree at least r.

We claim that every family of disjoint cycles in D' lifts to a family of the same size in D. Cycles avoiding z are unchanged. A loop at z is lifted to a 2-cycle uvu. For a longer cycle through z, replace z by the directed path uv if the entering arc originally terminated at u, and by v alone if it terminated at v. Since a disjoint family contains at most one cycle through z, these replacements preserve disjointness. Thus $\nu_L(D') \leq \nu(D) < k$, so D' is a counterexample of order smaller than $|D|$, contradicting the choice of D. Hence the endvertices of every arc have a common in-neighbour.

Finally, every vertex v has an in-neighbour, since otherwise deleting v would leave all remaining out-degrees unchanged and yield a smaller counterexample. For $u \in N_D^-(v)$, a common in-neighbour of u and v lies in $N_D^-(v)$ and points to u. Therefore the non-empty digraph $D[N_D^-(v)]$ has minimum in-degree at least one, and following incoming arcs until a vertex repeats produces a cycle. □

Proposition 4.3. *If a counterexample exists, one of minimum order has at most $(k-1)(r^2 - r + 1)$ vertices. In particular, when $r = 3k$, its order is less than $9k^3$.*

Proof. Choose D as in Lemma 4.2, and put $n = |V(D)|$. Construct an undirected graph J on $V(D)$ by joining distinct vertices u and v when $N^-(u) \cap N^-(v) \neq \varnothing$. An independent set in J corresponds to pairwise disjoint in-neighbourhoods in D. Each of these in-neighbourhoods contains a cycle, so the independence number of J satisfies $\alpha(J) \leq \nu(D) \leq k - 1$.

Every vertex $w \in V(D)$ has out-degree r, so w lies in exactly r of the sets $N_D^-(v)$. For any two of these in-neighbourhoods, their intersection contains w, and hence the corresponding two vertices of D are adjacent in J. Thus w accounts for at most $\binom{r}{2}$ edges of J. Summing over all n vertices of D, the total number of incidences between vertices of D and edges of J is at most $n\binom{r}{2}$. Every edge of J is counted at least once in this sum, so $e(J) \leq n\binom{r}{2}$, equivalently $2e(J) \leq nr(r-1)$.

Observe that the complement $\bar{J}$ of J contains no complete graph on $\alpha(J) + 1$ vertices, so Turán's theorem gives $e(\bar{J}) \leq (1 - 1/\alpha(J))n^2/2$. Since $e(J) + e(\bar{J}) = \binom{n}{2}$, it follows that $\alpha(J) \geq n^2/(n + 2e(J))$. Hence

$$k - 1 \geq \alpha(J) \geq \frac{n^2}{n + 2e(J)} \geq \frac{n}{r^2 - r + 1}.$$

This proves the claimed bound, and $n < 9k^3$ when $r = 3k$. □

5 Successive deletion of cycles

We first prove $\nu^*(G) \geq \delta^+(G)/2$. The key step is to find a non-empty proper part of the digraph whose boundary is small compared with the weight outside it. The weighted ball-growing argument we use is a directed, exact analogue of the region-growing technique of Garg, Vazirani and Yannakakis [10]; the directed version follows the spreading metric framework of Even, Naor, Rao and Schieber [9].

Lemma 5.1. *Let G be a strong oriented graph with at least two vertices, and let y be a fractional feedback vertex set with $0 < y_v < 1/2$ for every vertex. There are disjoint sets $U, S \subseteq V(G)$ such that U is non-empty and proper, every out-neighbour of a vertex of U lies in $U \cup S$, and $|S| \leq 2y(V(G)\setminus U) < |V(G)\setminus U|$.*

Proof. Give each arc vw length y_v. For a root a, let $d(a,v)$ be the shortest directed distance from a to v, with $d(a,a) = 0$; these distances are finite because G is strong. Every nontrivial closed directed walk has length at least one, since it contains a cycle whose weight is at least one.

For each root a, set $B_a = \{v : d(a,v) < 1/2\}$. We claim that $v \in B_a \setminus \{a\}$ implies $a \notin B_v$. Indeed, otherwise two paths of length $< 1/2$ would concatenate to a closed directed walk of length < 1, contradicting the fact that y is a fractional feedback vertex set.

Switching the order of summation and using the mutual exclusion,

$$\begin{aligned}\sum_a y_a \sum_{v \in B_a \setminus \{a\}} y_v(1-2y_v) &= \sum_v y_v(1-2y_v) \sum_{\substack{a:\, v \in B_a \\ a \neq v}} y_a \\ &\leq \sum_v y_v(1-2y_v)\, y(V(G) \setminus B_v) \\ &= \sum_a y_a(1-2y_a)\, y(V(G) \setminus B_a).\end{aligned}$$

Since $y_a > 0$ for every a, some root a satisfies

$$\sum_{v \in B_a \setminus \{a\}} y_v(1-2y_v) \leq (1-2y_a)\, y(V(G) \setminus B_a). \tag{5.1}$$

Fix this root a. Since G is strong and has at least two vertices, the root a has an out-neighbour w. The arc aw has length $y_a < 1/2$, so $d(a,w) \leq y_a < 1/2$, and hence $w \in B_a \setminus \{a\}$. Therefore the left-hand side in (5.1) is positive, and the chosen inequality implies $y(B_a) < y(V(G))$.

For $y_a < t < 1/2$, define $B(t) = \{v : d(a,v) \leq t\}$, $U(t) = \{v : d(a,v) + y_v \leq t\}$ and $S(t) = \{v : d(a,v) \leq t < d(a,v) + y_v\}$. Clearly, $B(t) = U(t) \cup S(t)$. For the root a, $d(a,a) = 0 \leq t$, so $a \in B(t)$; but $t > y_a = d(a,a) + y_a$, so $a \notin S(t)$. Hence $a \in U(t)$. If $v \in U(t)$ and vw is an arc, then $d(a,w) \leq d(a,v) + y_v \leq t$, so every out-neighbour of $U(t)$ lies in $B(t)$.

Set $g(t) = |S(t)| - 2y(S(t))$. We compute the average of $g(t)$ over $t \in (y_a, 1/2)$. As noted, the root a never belongs to $S(t)$. For any other vertex $v \in B_a \setminus \{a\}$, we have $d(a,v) < 1/2$, and $v \in S(t)$ exactly when

$$d(a,v) \leq t < d(a,v) + y_v.$$

This interval has length y_v. Throughout it, v is counted once in $|S(t)|$ and its weight y_v is counted once in $y(S(t))$, so its instantaneous contribution to $g(t) = |S(t)| - 2y(S(t))$ is $1 - 2y_v$. Over the whole interval, v therefore contributes $y_v(1-2y_v)$ to the total variation of $g(t)$. Vertices outside B_a never belong to $S(t)$, since their distance from a is at least $1/2$. By (5.1), the average of $g(t)$ over $t \in (y_a, 1/2)$ is

$$\frac{1}{1/2 - y_a} \sum_{v \in B_a \setminus \{a\}} y_v(1-2y_v) \leq 2\, y(V(G) \setminus B_a).$$

Since g is piecewise constant, choose t with $g(t) \leq 2\, y(V(G) \setminus B_a)$. As $B(t) \subseteq B_a$, we have $y(V(G) \setminus B(t)) \geq y(V(G) \setminus B_a)$, and therefore

$$|S(t)| - 2y(S(t)) = g(t) \leq 2y(V(G) \setminus B(t)).$$

Since $V(G) \setminus U(t) = S(t) \cup (V(G) \setminus B(t))$, this gives $|S(t)| \leq 2y(V(G) \setminus U(t))$. Take $U = U(t)$ and $S = S(t)$. Then $a \in U$, so U is non-empty, and $y(U) \leq y(B_a) < y(V(G))$, so U is proper. All outgoing arcs of U end in $U \cup S$. □

Theorem 5.2. *Every digraph G satisfies $\nu^*(G) \geq \delta^+(G)/2$.*

Proof. Let $\delta = \delta^+(G)$. Suppose that there exists a digraph G such that $\nu^*(G) < \delta/2$ and choose a counterexample G of minimum order. Then $\delta \geq 1$ and thus G has a cycle. By (3.1), we may choose a fractional feedback vertex set y with $y(V(G)) = \nu^*(G) < \delta/2$. Since y gives weight at least one to every cycle, $y(V(G)) \geq 1$, and hence $\delta \geq 3$.

The digraph G is strong. Indeed, a terminal strong component H of G has no out-going arcs to the rest of G and hence has minimum out-degree at least δ. Thus the minimality of G and the fact that $y(V(H)) \leq y(V(G))$ implies that $G = H$.

First suppose that $y_v \geq 1/2$ for some vertex v. Then $G-v$ has minimum out-degree at least $\delta-1$, and the restricted weights have total $y(V(G))-y_v < (\delta-1)/2$. Thus $G-v$ is again a smaller counterexample, contradiction. Thus we must have $y_v < 1/2$ for every vertex $v \in V(G)$. In particular, G has no 2-cycle, since the weights on such a cycle would sum to less than one, contradicting that y is a fractional vertex cover.

Next suppose that $y_v = 0$ for some vertex v. We delete v and, for every pair of arcs uv and vw in G, add the arc uw; if several pairs give the same arc, we keep only one copy. No loop is created in this process, because G has no 2-cycle. The resulting digraph still has minimum out-degree at least δ. The restricted weights remain a fractional feedback vertex set: replacing each new arc uw on a cycle by the two-arc path $u \to v \to w$ gives a closed directed walk of the same weight in G, since $y_v = 0$. Thus the resulting smaller digraph is therefore a counterexample, a contradiction.

We now have $0 < y_v < 1/2$ for every vertex, so Lemma 5.1 gives disjoint non-empty proper sets $U, S \subseteq V(G)$ such that every out-neighbour of a vertex of U lies in $U \cup S$ and $|S| \leq 2y(V(G) \setminus U)$. Since U is a proper subset of $V(G)$, the minimality of G gives $\nu^*(G[U]) \geq \delta^+(G[U])/2$. The restriction of y to U is a feasible fractional feedback vertex set on $G[U]$, so $\nu^*(G[U]) \leq y(U)$. Moreover, every out-neighbour of U lies in $U \cup S$, so deleting S removes at most $|S|$ out-neighbours at any vertex of U, giving $\delta^+(G[U]) \geq \delta - |S|$. Combining these estimates,

$$\begin{aligned}\delta \leq \delta^+(G[U]) + |S| &\leq 2\nu^*(G[U]) + |S| \\ &\leq 2y(U) + |S| \\ &\leq 2y(U) + 2y(V(G) \setminus U) \\ &= 2y(V(G)) < \delta,\end{aligned}$$

a contradiction. This completes the proof. □

Let H be a digraph. If all vertices of a cycle C are removed from H, a remaining vertex v loses exactly $|N_H^+(v) \cap V(C)|$ out-neighbours. We seek a probability distribution on the cycles of H that bounds both the expectation and the largest possible value of this loss, for every vertex. Fix an integer $n \geq |V(H)| \geq 2$, and put $\ell = \log(2n)$.

Under the minimum out-degree conditions considered in this work, Lemmas 5.3-5.5 allow us to extract an integer disjoint cycle packing whose size is at least $\nu^*(G) - o(k)$ from a fractional cycle packing of weight $\nu^*(G)$.

Lemma 5.3. *Suppose that $0 < \tau \leq 1/4$ and $24\tau^{-2}\ell \leq s \leq \delta^+(H)$. There is a probability distribution on the cycles of H such that, writing C^* for a cycle drawn from this distribution, every vertex $v \in V(H)$ satisfies*

$$\mathbb{P}[v \in V(C^*)] \leq \frac{2}{(1-\tau)s} \quad \text{and} \quad \mathbb{E}\big[|N_H^+(v) \cap V(C^*)|\big] \leq \frac{2d_H^+(v)}{(1-\tau)s}.$$

Moreover, with positive probability every cycle C^* satisfies

$$|N_H^+(v) \cap V(C^*)| \leq \frac{24(1+\tau)\ell\, d_H^+(v)}{\tau^2 s}$$

for every vertex v.

Proof. Put $t = \lfloor \tau^2 s/(12\ell) \rfloor \geq 2$, and colour the vertices of H independently and uniformly with t colours. For a fixed vertex v and colour i, let X be the number of out-neighbours of v of colour i. Then X is binomialy distributed with expected value $\mu = d_H^+(v)/t \geq 12\ell/\tau^2$, so the Chernoff bound gives

$$\mathbb{P}\big[|X - \mu| > \tau\mu\big] \leq 2e^{-\tau^2\mu/3} \leq 2e^{-4\ell}.$$

There are at most $nt \leq n^2$ pairs (v, i), since $t \leq s \leq |V(H)| - 1 < n$. Hence, by the union bound the probability that some bound fails is at most $2n^2e^{-4\ell} = 1/(8n^2) < 1$. We may therefore fix a colouring in

which every vertex has between $(1-\tau)d_H^+(v)/t$ and $(1+\tau)d_H^+(v)/t$ out-neighbours in each colour class $V_1,\dots,V_t$. In particular, every colour class is non-empty and $\delta^+(H[V_i]) \geq (1-\tau)s/t$ for all $i \in [t]$.

By Theorem 5.2, each $H[V_i]$ has a fractional cycle packing $\{x_C\}$ with total weight

$$\nu_i^* = \sum_{C \subseteq V_i} x_C \geq \frac{(1-\tau)s}{2t}.$$

Choose I uniformly at random from $\{1,\dots,t\}$. Conditional on $I=i$, select a cycle C in $H[V_i]$ with probability x_C/ν_i^*. Let C^* denote the cycle thus selected. For a cycle $C \subseteq V_i$, the probability that $C^* = C$ is $x_C/(t\nu_i^*)$. Consider an arbitrary vertex u and let $j \in [t]$ be the index so that $u \in V_j$. If $I \neq j$, then $u \notin V(C^*)$; if $I = j$, then $u \in V(C^*)$ exactly when the selected cycle in $H[V_j]$ contains u. Hence

$$\mathbb{P}[u \in V(C^*)] = \frac{1}{t\nu_i^*} \sum_{C \ni u} x_C \leq \frac{1}{t\nu_i^*} \leq \frac{2}{(1-\tau)s},$$

where the inequality uses the capacity constraint $\sum_{C \ni u} x_C \leq 1$ and $\nu_i^* \geq (1-\tau)s/(2t)$. Summing over $u \in N_H^+(v)$,

$$\mathbb{E}\big[|N_H^+(v) \cap V(C^*)|\big] = \sum_{u \in N_H^+(v)} \mathbb{P}[u \in V(C^*)] \leq \frac{2d_H^+(v)}{(1-\tau)s}.$$

Finally, C^* lies in a single colour class V_i, and therefore

$$|N_H^+(v) \cap V(C^*)| \leq |N_H^+(v) \cap V_i| \leq \frac{(1+\tau)d_H^+(v)}{t} \leq \frac{24(1+\tau)\ell\, d_H^+(v)}{\tau^2 s},$$

where the last inequality uses $t = \lfloor \tau^2 s/(12\ell) \rfloor \geq \tau^2 s/(24\ell)$. □

The next lemma shows that, after deleting one cycle, the q-th power of the reciprocal out-degree of every vertex does not increase in expectation. This will be applied at each step of the deletion sequence in Lemma 5.5.

Lemma 5.4. *Let $q \geq 1$ be an integer, let $0 < a < b < s$, and suppose that $\delta^+(H) \geq s$. Let C^* be a random cycle such that, for every vertex v,*

$$\mathbb{E}\big[|N_H^+(v) \cap V(C^*)|\big] \leq \frac{a\, d_H^+(v)}{s} \text{ and } |N_H^+(v) \cap V(C^*)| \leq \rho\, d_H^+(v).$$

If $(q+1)\rho \leq 1 - a/b$, then $H - V(C^*)$ is non-empty and every vertex outside C^* has positive out-degree in $H - V(C^*)$. Moreover, for every vertex v,

$$\mathbb{E}\left[\left(\frac{s-b}{d^+_{H-V(C^*)}(v)}\right)^q\right] \leq \left(\frac{s}{d_H^+(v)}\right)^q.$$

Proof. The hypothesis gives

$$\frac{|N_H^+(v) \cap V(C^*)|}{d_H^+(v)} \leq \rho < \frac{1}{q+1} < 1,$$

so $d^+_{H-V(C^*)}(v) > 0$ for every $v \notin V(C^*)$, and $H - V(C^*)$ is nonempty.

For $0 \leq x < 1/(q+1)$, Lemma 3.4 gives $(1-x)^{-q} \leq 1 + qx/(1-(q+1)x)$. Substitute $x = |N_H^+(v) \cap V(C^*)|/d_H^+(v) \leq \rho \leq 1/(q+1)$. Using $1-(q+1)\rho \geq a/b$,

$$\left(\frac{d_H^+(v)}{d_H^+(v) - |N_H^+(v) \cap V(C^*)|}\right)^q \leq 1 + \frac{qb}{a\, d_H^+(v)} |N_H^+(v) \cap V(C^*)|.$$

Taking expectations over the choice of C^* and using $\mathbb{E}[|N_H^+(v) \cap V(C^*)|] \leq a\, d_H^+(v)/s$,

$$\mathbb{E}\left[\left(\frac{d_H^+(v)}{d_H^+(v) - |N_H^+(v) \cap V(C^*)|}\right)^q\right] \leq 1 + \frac{qb}{s} \leq e^{qb/s}.$$

Since $(1-b/s)^q \le e^{-qb/s}$, multiplying both sides by $(1-b/s)^q (s/d_H^+(v))^q$ and using that $d^+_{H-V(C^*)}(v) = d_H^+(v) - |N_H^+(v) \cap V(C^*)|$ yields

$$\mathbb{E}\left[\left(\frac{s-b}{d^+_{H-V(C^*)}(v)}\right)^q\right] \le \left(\frac{s}{d_H^+(v)}\right)^q,$$

which completes the proof. □

We now prescribe a decreasing sequence $\{s_i\}$ of out-degree bounds. The decrease at each step approaches two when the current bound is large. A probability estimate for each vertex shows that there is a sequence of cycle deletions satisfying all the bounds.

Lemma 5.5. *Let $n \ge 2$ be an integer, and let D be a digraph with $|V(D)| \le n$ and minimum out-degree $r \ge 1$. Put $\ell = \log(2n)$. Fix an integer $q > \ell$, and set $s_0 = r(1-\ell/q)$, $\kappa = 30(q+1)\ell$, and $s_* = 64\kappa$. If $s_0 \ge s_*$, then D contains a family of m disjoint cycles with $m \ge s_0/2 - (3/2)\kappa^{1/3} s_0^{2/3} - 8\kappa$. Their deletion leaves a non-empty induced subdigraph with minimum out-degree greater than $s_* - 4$.*

Proof. Set $\tau_j = (\kappa/s_j)^{1/3}$ and $b_j = 2/(1-2\tau_j)$. Starting from s_0, set $s_{j+1} = s_j - b_j$ until the first m with $s_m < s_*$. For every $j < m$, $s_j \ge s_* = 64\kappa$, and hence $\tau_j \le 1/4$ and $2 \le b_j \le 4$. Thus m is finite. The final digraph has minimum out-degree greater than $s_m \ge s_* - 4$.

At step j, the current digraph H_j has minimum out-degree greater than s_j. Since $s_j \ge 64\kappa$, we have $s_j^{1/3} \ge 4\kappa^{1/3}$ and thus

$$24\tau_j^{-2}\ell = \frac{24\ell\, s_j}{s_j^{1/3}\kappa^{2/3}} \le \frac{6\ell}{\kappa} s_j \le s_j,$$

so Lemma 5.3 with parameter τ_j gives a probability distribution on the cycles of H_j. It satisfies

$$\mathbb{E}[|N^+_{H_j}(v) \cap V(C^*)|] \le \frac{2d^+_{H_j}(v)}{(1-\tau_j)s_j}, \qquad |N^+_{H_j}(v) \cap V(C^*)| \le \frac{24(1+\tau_j)\ell\, d^+_{H_j}(v)}{\tau_j^2 s_j}.$$

These are the hypotheses of Lemma 5.4 with $a = 2/(1-\tau_j)$, $b = b_j$, and $\rho = 24(1+\tau_j)\ell/(\tau_j^2 s_j)$. Note that $0 < a < b_j \le 4 < s_j$. Moreover,

$$(q+1)\rho = \frac{24(1+\tau_j)(q+1)\ell}{\tau_j^2 s_j} \le \frac{30(q+1)\ell}{\tau_j^2 s_j} = \tau_j < 1 - \frac{a}{b_j}.$$

Thus Lemma 5.4 applies at every step.

We now choose m successive cycles at random. Start with $H_0 = D$. At step $j < m$, choose a cycle C_{j+1} from the distribution above and set $H_{j+1} = H_j - V(C_{j+1})$. If any remaining vertex has out-degree at most s_{j+1} in H_{j+1}, stop; otherwise continue. For each vertex v, define the random variable $Z_j(v)$ is follows

$$Z_j(v) = \begin{cases} 0, & v \text{ has been removed,} \\ (s_j/d^+_{H_j}(v))^q, & \text{otherwise.} \end{cases}$$

Lemma 5.4 gives $\mathbb{E}[Z_{j+1}(v)] \le Z_j(v)$ at every step. Iterating,

$$\mathbb{E}[Z_m(v)] \le Z_0(v) = \left(\frac{s_0}{r}\right)^q = \left(1 - \frac{\ell}{q}\right)^q < 1.$$

If the process stops early, some remaining vertex v has out-degree at most the current threshold, so $Z_m(v) \ge 1$. This holds because $d^+_{H_m}(v) \le s_m$, hence $s_m/d^+_{H_m}(v) \ge 1$. Since $Z_m(v) \ge 0$ and $\mathbb{E}[Z_m(v)] \le (1-\ell/q)^q$, Markov's inequality gives

$$\mathbb{P}[Z_m(v) \ge 1] \le \mathbb{E}[Z_m(v)] \le \left(1 - \frac{\ell}{q}\right)^q.$$

Taking a union bound over all vertices, the probability of stopping early is at most

$$n\left(1 - \frac{\ell}{q}\right)^q \le ne^{-\ell} = \frac{1}{2}.$$

Therefore, with positive probability all m deletions succeed. Choosing such a sequence of cycles gives m disjoint cycles, and the remaining digraph has minimum out-degree greater than $s_m \geq s_* - 4$.

It remains to estimate m. Since $s_j - s_{j+1} = b_j$,

$$m = \sum_{j=0}^{m-1} 1 = \sum_{j=0}^{m-1} \frac{s_j - s_{j+1}}{b_j}.$$

Note that $1/b_j = 1/2 - (\kappa/s_j)^{1/3}$, and this function of s_j is increasing. For all $u \in [s_{j+1}, s_j]$ we have

$$\frac{1}{2} - \left(\frac{\kappa}{u}\right)^{1/3} \leq \frac{1}{b_j}.$$

Therefore

$$\frac{s_j - s_{j+1}}{b_j} \geq \int_{s_{j+1}}^{s_j} \left(\frac{1}{2} - \left(\frac{\kappa}{u}\right)^{1/3}\right) du.$$

Summing over j, the intervals cover $[s_*, s_0]$, and

$$m \geq \int_{s_*}^{s_0} \left(\frac{1}{2} - \left(\frac{\kappa}{u}\right)^{1/3}\right) du = \frac{s_0}{2} - \frac{3}{2}\kappa^{1/3} s_0^{2/3} - 8\kappa,$$

using $s_* = 64\kappa$. □

Theorem 5.6. *Every digraph D of order n and minimum out-degree $r \geq 1$ satisfies $\nu(D) \geq \max\{0, r/2 - 7r^{3/4}\sqrt{\log(2n)}\}$.*

Proof. Put $\ell = \log(2n)$ and choose $q = \lceil r^{1/4}\sqrt{\ell} \rceil$. If $r/2 < 7r^{3/4}\sqrt{\ell}$, then $\nu(D) \geq 0$ already gives the result. We may therefore assume $r \geq 14r^{3/4}\sqrt{\ell}$. This implies that $r^{1/4} \geq 14\sqrt{\ell}$ so $q > \ell$ holds.

Set $s_0 = r(1-\ell/q)$ and $\kappa = 30(q+1)\ell$. Then $s_0 \geq r - r^{3/4}\sqrt{\ell} > 13r/14$. Since $q \leq r^{1/4}\sqrt{\ell}+1$, we have $\kappa = 30(q+1)\ell \leq 30(r^{1/4}\sqrt{\ell}+2)\ell < 35r^{1/4}\ell^{3/2}$. Our assumption $r \geq 14r^{3/4}\sqrt{\ell}$ ensures $64\kappa < s_0$, because $s_0 \geq 13r/14$ and $64\kappa = 1920(q+1)\ell$ is dominated by r under our parameter bound. Thus $s_* = 64\kappa < s_0$, so Lemma 5.5 applies and gives

$$\nu(D) \geq \frac{s_0}{2} - \frac{3}{2}\kappa^{1/3} s_0^{2/3} - 8\kappa.$$

The two error terms are at most $5r^{3/4}\sqrt{\ell}$ and $\frac{10}{7}r^{3/4}\sqrt{\ell}$, respectively. Hence $\nu(D) \geq \frac{r}{2} - 7r^{3/4}\sqrt{\ell}$. □

6 Proof of the main theorem

We are now ready to prove the main theorem which we restate below.

Theorem 1.2. *There exist absolute constants $C > 0$ and K such that, for all integers $k \geq K$ and $n \geq 1$, every n-vertex digraph D with*

$$\delta^+(D) \geq 2k + Ck^{3/4}\sqrt{\log k}$$

contains k disjoint directed cycles.

Proof. We establish the stronger explicit bound

$$f(k) \leq \left\lceil 2k + 64k^{3/4}\sqrt{\log(4k)} \right\rceil \quad \text{for } k \geq 10^{12}.$$

Since $\log(4k) \leq 2\log k$ for $k \geq 4$, this implies Theorem 1.2 with $C = 64\sqrt{2}$ and $K = 10^{12}$.

Set $E_k = k^{3/4}\sqrt{\log(4k)}$ and $r = \lceil 2k + 64E_k \rceil$. The function $x^{-1/4}\sqrt{\log(4x)}$ is decreasing for $x \geq 10^{12}$, so

$$\frac{64E_k}{k} \leq 64 \cdot 10^{-3}\sqrt{\log(4 \cdot 10^{12})} < \frac{48}{125} < 1,$$

and hence $2k \leq r < 3k$.

Suppose there is a counterexample at (r, k), i.e. a digraph D with $\delta^+(D) \geq r$ and $\nu(D) < k$. By Lemma 4.2, take one of minimum order n that is r-out-regular. Proposition 4.3 gives $n < 9k^3$, so $\log(2n) < 3\log(4k)$.

Theorem 5.6 yields

$$\nu(D) \geq \frac{r}{2} - 7r^{3/4}\sqrt{\log(2n)} > k,$$

where the strict inequality follows because $r \geq 2k + 64E_k$, while $r < 3k$ and $\log(2n) < 3\log(4k)$ give $7r^{3/4}\sqrt{\log(2n)} \leq 7 \cdot 3^{5/4}E_k < 32E_k$, a contradiction.

This shows that no counterexample at (r, k) exists. Hence every digraph with minimum out-degree at least r contains k disjoint cycles, so $f(k) \leq r = \lceil 2k + 64E_k \rceil$, as claimed. □

Use of AI-assisted tools

Following Lemma 5.4, we establish the $o(k)$ bound using martingale methods. We further attempted to obtain an explicit linear bound of the form $2k + C$ for some absolute constant C, but this attempt was unsuccessful. Nevertheless, AI-assisted refinement allowed us to improve the asymptotic error term from $2k + o(k)$ to the explicit bound $O(k^{3/4}\sqrt{\log k})$. Additionally, AI tools aided the construction of the sequence $Z_j(v)$ in Lemma 5.5: specifically, removed vertices are assigned a value of zero, and all values are frozen after the first unsuccessful deletion operation.

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